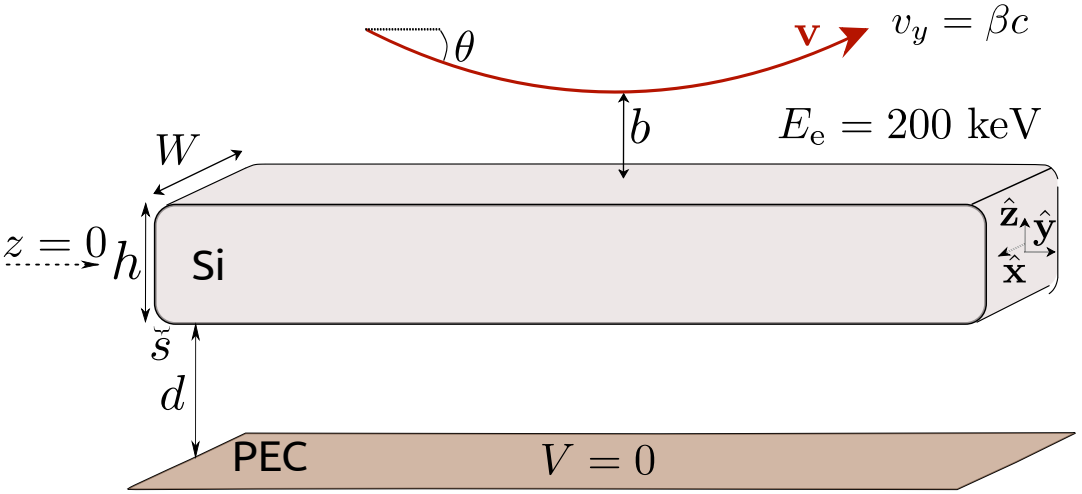


Electron-coupling-to-waveguide

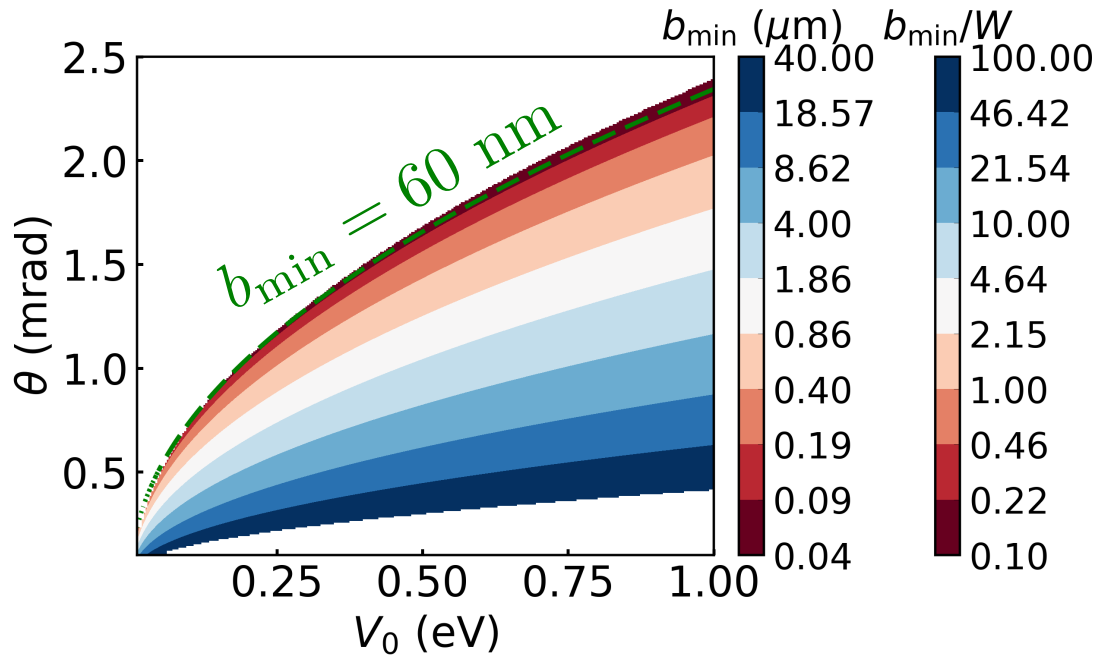
Correction of the trajectory and integration

$$\Gamma_{\text{EELS}}(\omega) = 2 \int_{z_{\min}}^{\infty} \frac{v dz}{\sqrt{\left| c^2 \beta^2 \sin^2 \theta + \frac{2eV(z)}{m_e \gamma_e} \right|}} \frac{d\Gamma_{\text{EELS}}(\omega, z)}{dy}$$

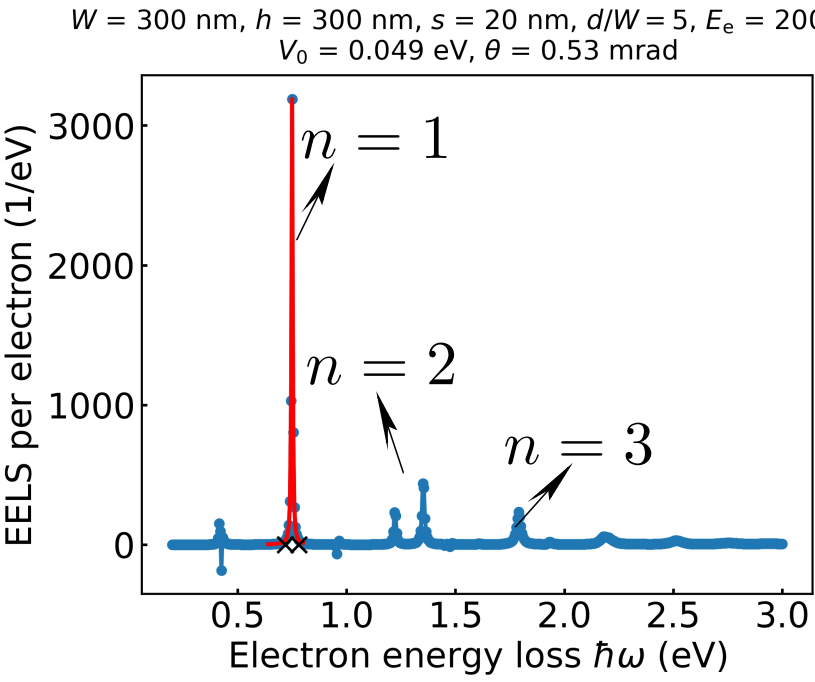
$W = 400 \text{ nm}$
 $h = 300 \mu\text{m}$



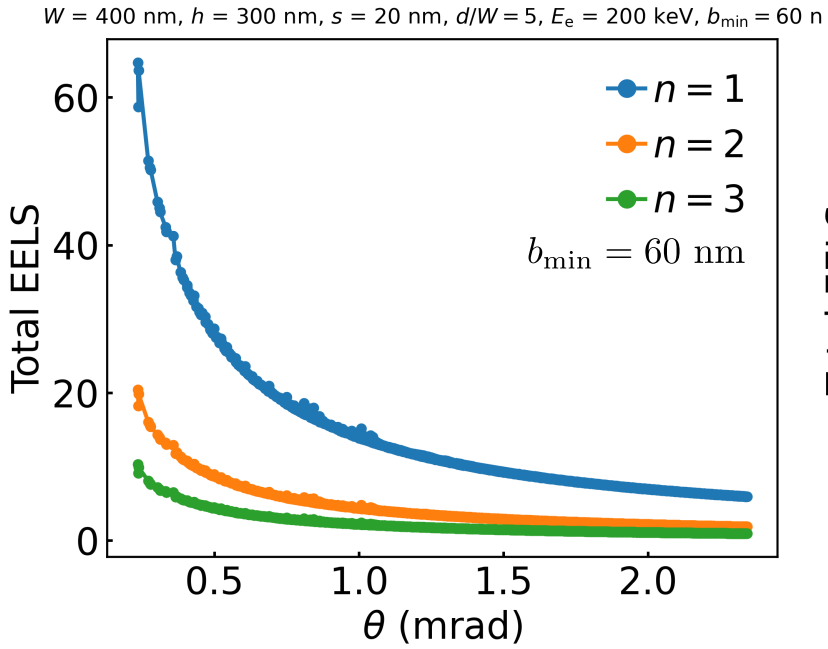
$d = 2 \mu\text{m}$



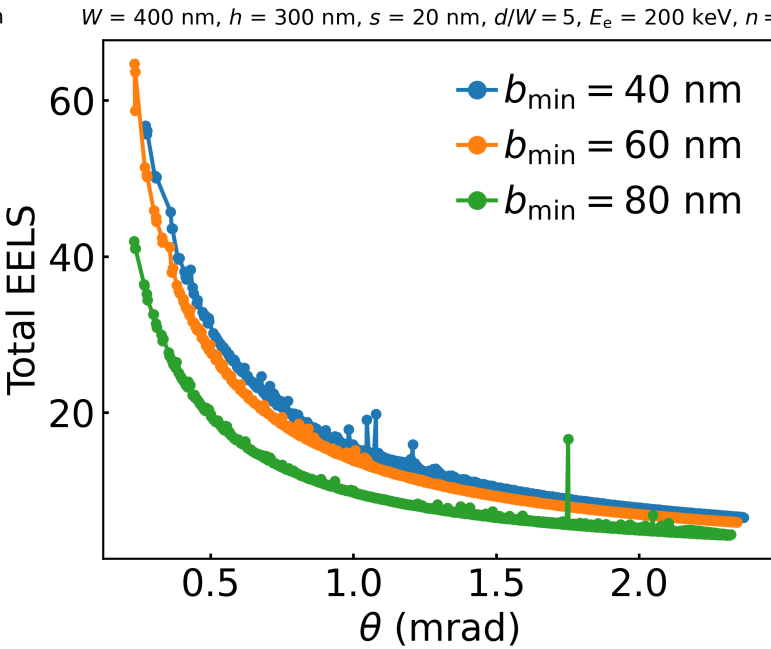
The position of peaks depends on width and height



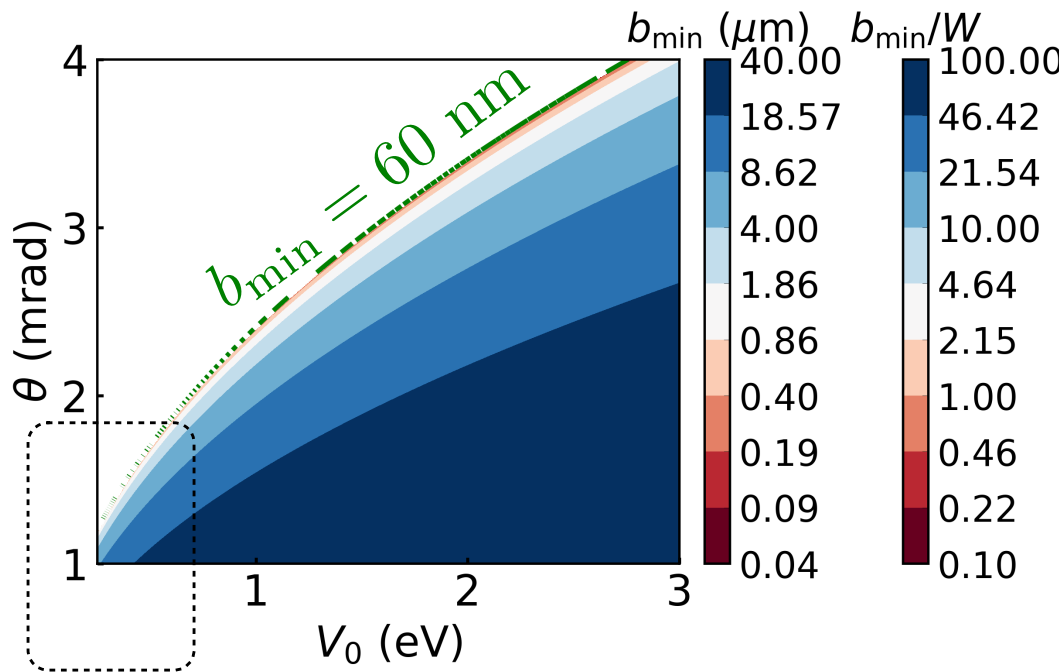
EELS for different modes for fixed b



EELS for different b for n = 1



$d = 100 \mu\text{m}$



For bigger d , the angles are higher

